Database Systems	
Lab #:	**10**
Lab Title:	Implementation of different types of operators in SQL
Submitted by:	

Names	Registration #
Sumaira Safeer	FA22-BCE-019

Rubrics name & number		Marks	
		In-Lab	Post-Lab
Engineering Knowledge	***R2: Use of Engineering Knowledge and follow Experiment Procedures:*** *Ability to follow experimental procedures, control variables, and record procedural steps on lab report.*		
Problem Analysis	***R6: Experimental Data Analysis:*** *Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.*		
Design	***R8: Best Coding Standards:*** *Ability to follow the coding standards and programming practices.*		
Modern Tools Usage	***R9: Understand Tools:*** *Ability to describe and explain the principles behind and applicability of engineering tools.*		
Individual and Teamwork	***R12: Individual Work Contributions:*** *Ability to carry out individual responsibilities.*		
	R13: Management of Team Work: *Ability to appreciate, understand and work with multidisciplinary team members.*		

Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab No: 10

Title: Implementation of different types of operators in SQL.

- Arithmetic Operator.
- Logical Operator.
- Comparison Operator.
- Special Operator.
- Set Operator.

Objective:

To understand the working of SELECT queries and use different conditions involving logical and arithmetic operators to get the user desired results through queries.

In lab Tasks:

Design a table 'Customer'.

Table:

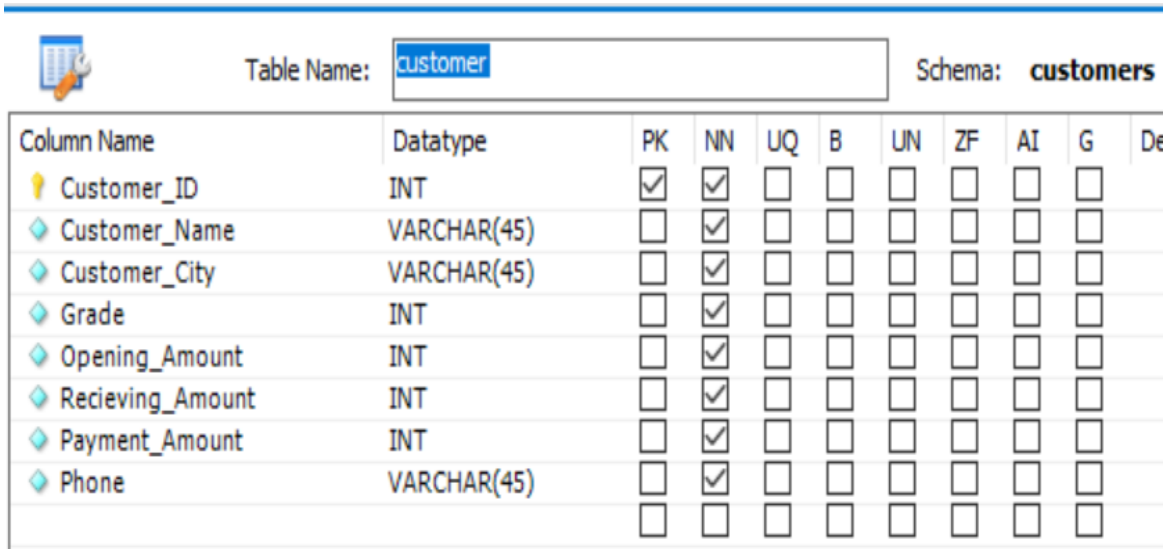
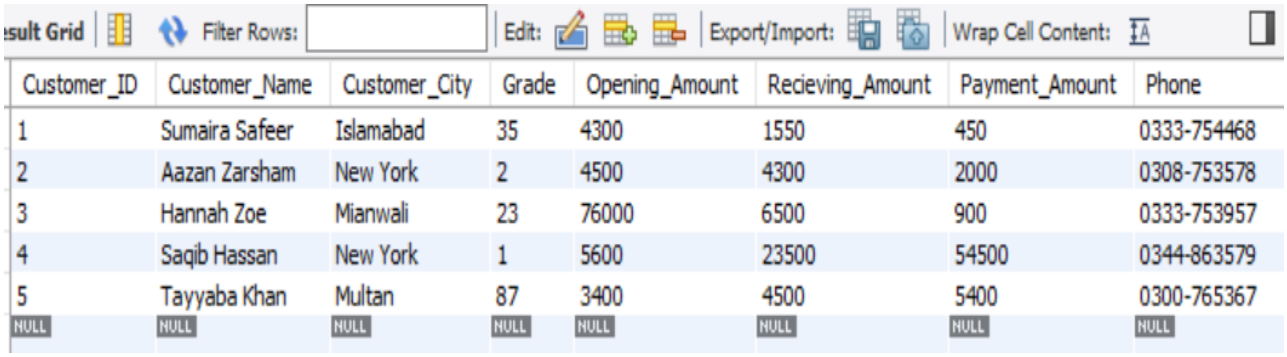


Table Name: customer Schema: customers

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	De
Customer_ID	INT	☑	☑	☐	☐	☐	☐	☐	☐	
Customer_Name	VARCHAR(45)	☐	☑	☐	☐	☐	☐	☐	☐	
Customer_City	VARCHAR(45)	☐	☑	☐	☐	☐	☐	☐	☐	
Grade	INT	☐	☑	☐	☐	☐	☐	☐	☐	
Opening_Amount	INT	☐	☑	☐	☐	☐	☐	☐	☐	
Recieving_Amount	INT	☐	☑	☐	☐	☐	☐	☐	☐	
Payment_Amount	INT	☐	☑	☐	☐	☐	☐	☐	☐	
Phone	VARCHAR(45)	☐	☑	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

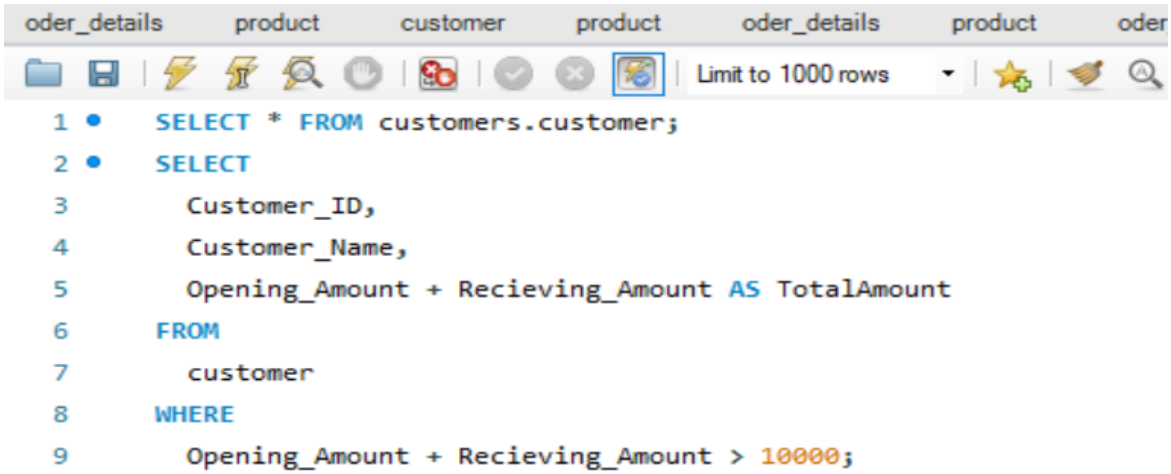
Insertion of data:



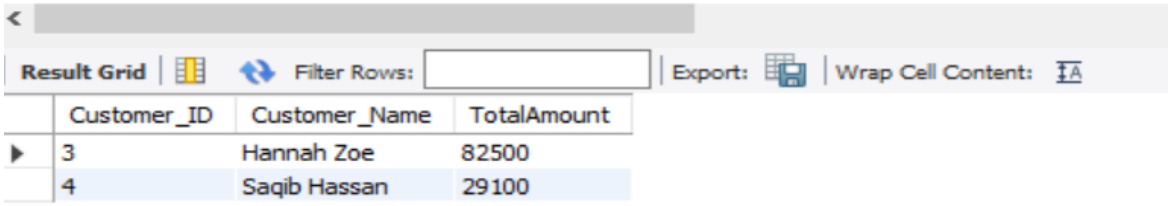
Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
1	Sumaira Safeer	Islamabad	35	4300	1550	450	0333-754468
2	Aazan Zarsham	New York	2	4500	4300	2000	0308-753578
3	Hannah Zoe	Mianwali	23	76000	6500	900	0333-753957
4	Saqib Hassan	New York	1	5600	23500	54500	0344-863579
5	Tayyaba Khan	Multan	87	3400	4500	5400	0300-765367
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Task1:

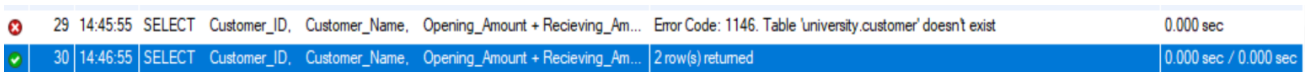
Display customer id customer name and summation of opening amount and receiving amount where this summation is greater than 10,000.



```
1  SELECT * FROM customers.customer;
2  SELECT
3    Customer_ID,
4    Customer_Name,
5    Opening_Amount + Recieving_Amount AS TotalAmount
6  FROM
7    customer
8  WHERE
9    Opening_Amount + Recieving_Amount > 10000;
```



Customer_ID	Customer_Name	TotalAmount
3	Hannah Zoe	82500
4	Saqib Hassan	29100



29	14:45:55	SELECT Customer_ID, Customer_Name, Opening_Amount + Recieving_Am...	Error Code: 1146. Table 'university.customer' doesn't exist	0.000 sec
30	14:46:55	SELECT Customer_ID, Customer_Name, Opening_Amount + Recieving_Am...	2 row(s) returned	0.000 sec / 0.000 sec

Task 2:

Display customer id and difference of opening amount and payment amount.

```
11
12
13 •      SELECT
14        Customer_ID,
15         Opening_Amount - Recieving_Amount AS Difference
16      FROM
17        customer;
18
19
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Difference
1	2750
2	200
3	69500
4	-17900
5	-1100

Task 3:

Display customer id, customer name, customer city and payment amount multiplied with 2.

```
18
19 •    SELECT
20        Customer_ID,
21        Customer_Name,
22        Customer_City,
23        Payment_Amount * 2 AS DoublePayment
24      FROM
25        customer;
26
```

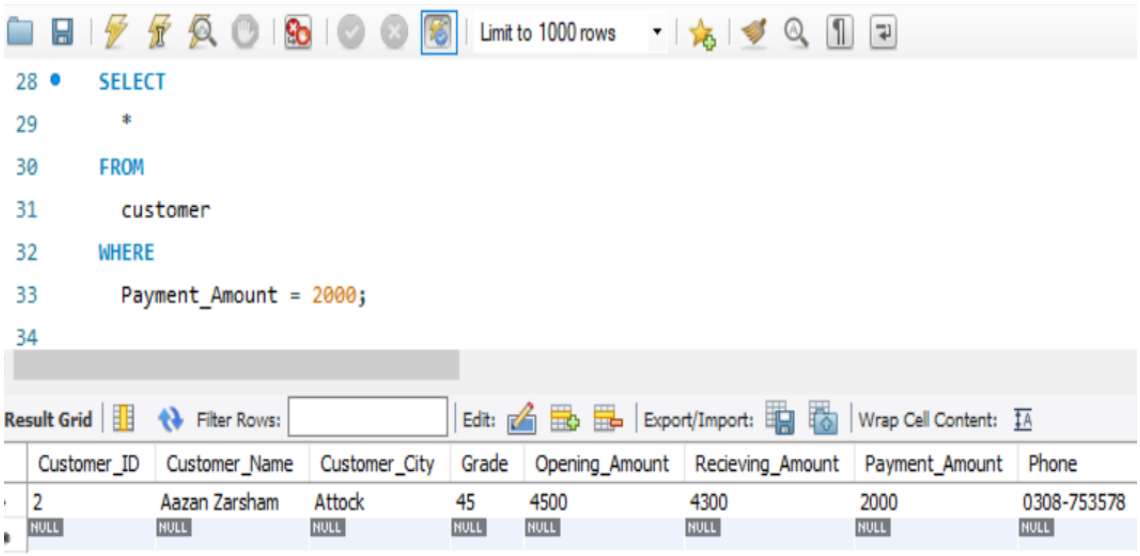
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	DoublePayment
1	Sumaira Safeer	Islamabad	46900
2	Aazan Zarsham	Attock	6800
3	Hannah Zoe	Mianwali	10800
4	Saqib Hassan	Lahore	109000
5	Tayyaba Khan	Multan	10800

32 | 14:57:43 | SELECT Customer_ID, Customer_Name, Customer_City, Payment_Amount... | 5 row(s) returned | 0.000 sec / 0.000 sec

Task 4:

Display all records of customer where payment amount is equal to 2000.



```
28 • SELECT
29     *
30   FROM
31     customer
32   WHERE
33     Payment_Amount = 2000;
34
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

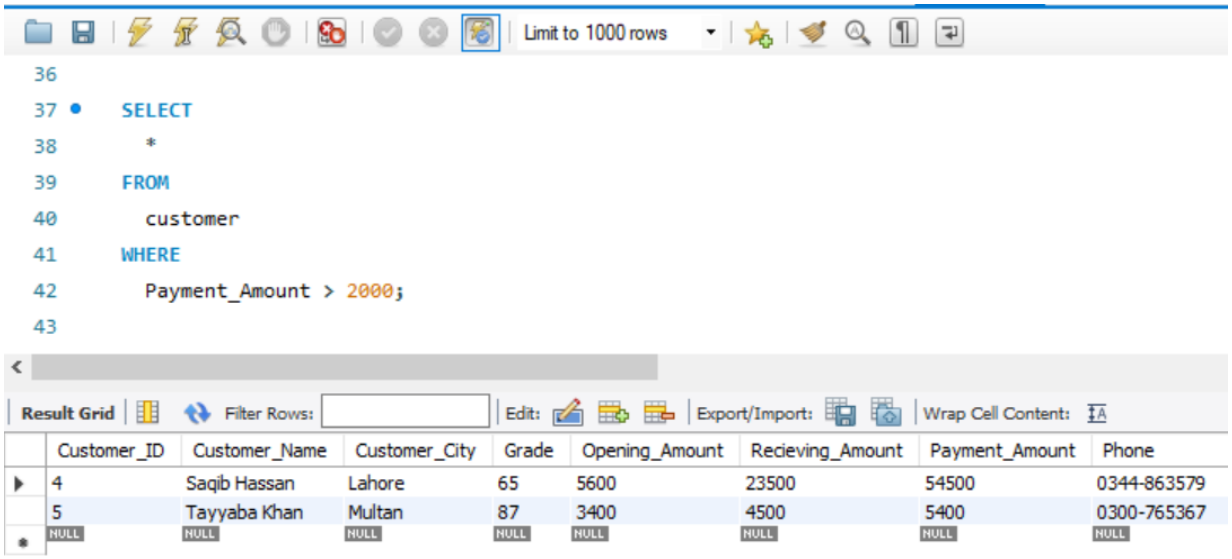
Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
2	Aazan Zarsham	Attock	45	4500	4300	2000	0308-753578
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

33 | 15:00:47 | SELECT * FROM customer WHERE Payment_Amount = 2000 LIMIT 0, 1000 | 0 row(s) returned | 0.000 sec / 0.000 sec

Task 5:

Display all records of customer where payment amount is greater than, less than, greater than and equal to, less than and equal to, not equal to 2000.

- **Greater than 2000:**

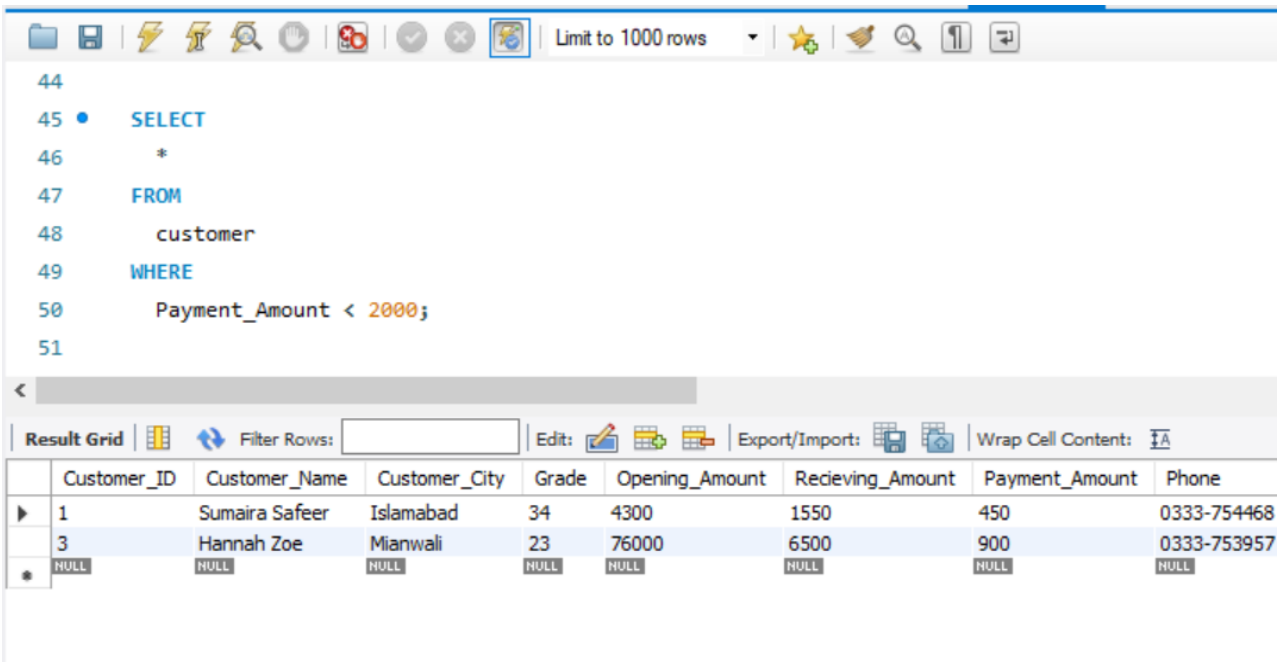


```
36
37 • SELECT
38     *
39   FROM
40     customer
41   WHERE
42     Payment_Amount > 2000;
43
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
4	Saqib Hassan	Lahore	65	5600	23500	54500	0344-863579
5	Tayyaba Khan	Multan	87	3400	4500	5400	0300-765367
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

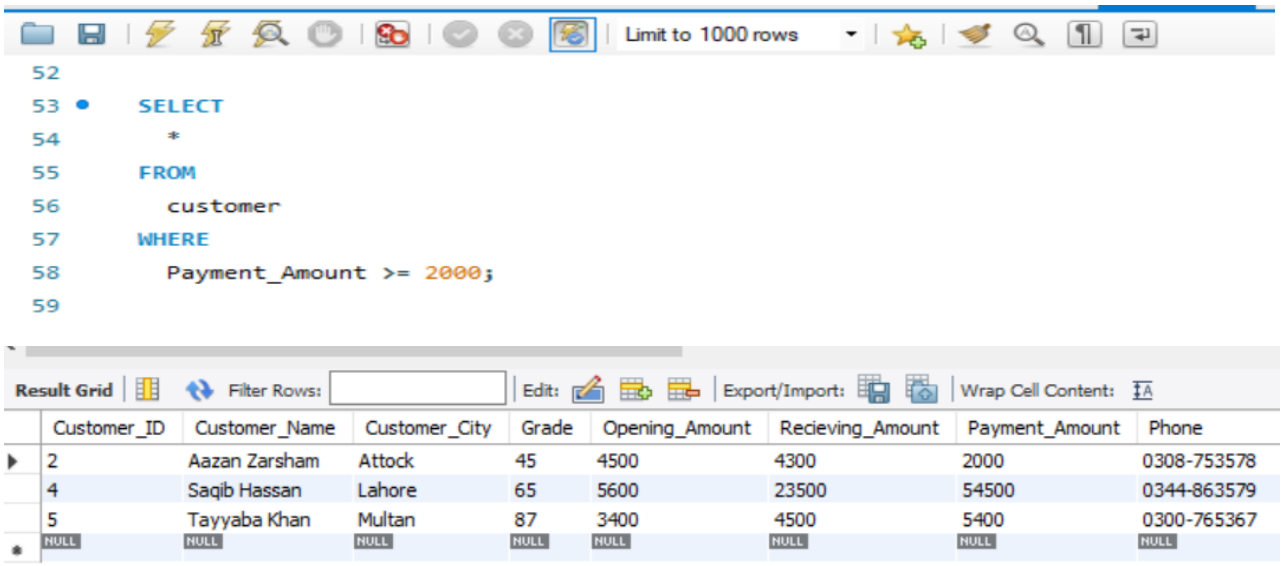
- **Less than 2000:**



```
SELECT
  *
FROM
  customer
WHERE
  Payment_Amount < 2000;
```

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
1	Sumaira Safeer	Islamabad	34	4300	1550	450	0333-754468
3	Hannah Zoe	Mianwali	23	76000	6500	900	0333-753957
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

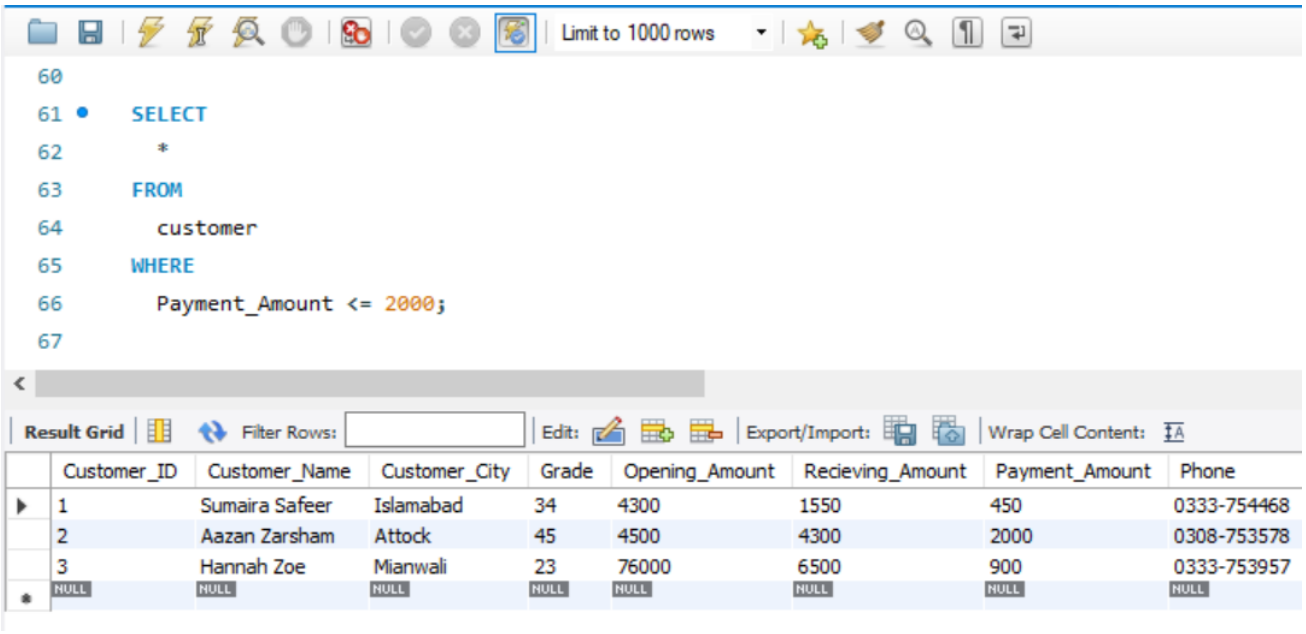
- **Greater than or equal to 2000:**



```
SELECT
  *
FROM
  customer
WHERE
  Payment_Amount >= 2000;
```

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
2	Aazan Zarsham	Attock	45	4500	4300	2000	0308-753578
4	Saqib Hassan	Lahore	65	5600	23500	54500	0344-863579
5	Tayyaba Khan	Multan	87	3400	4500	5400	0300-765367
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- **Less than or equal to 2000:**

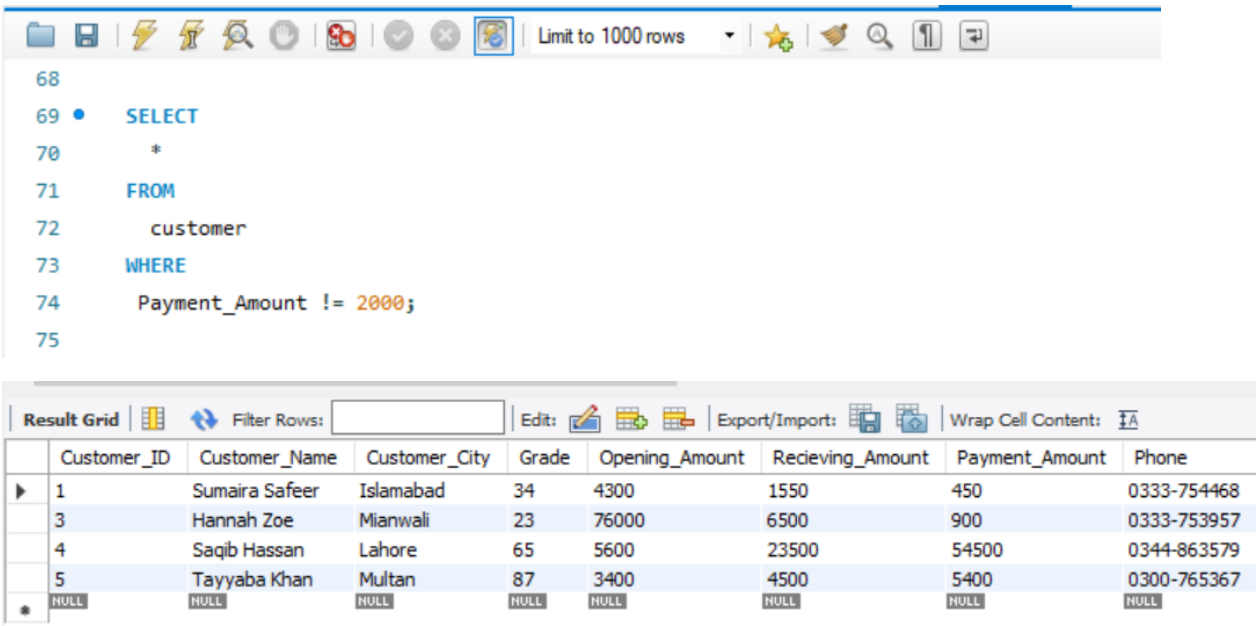


```
60
61 •  SELECT
62      *
63    FROM
64      customer
65    WHERE
66      Payment_Amount <= 2000;
67
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
1	Sumaira Safeer	Islamabad	34	4300	1550	450	0333-754468
2	Aazan Zarsham	Attock	45	4500	4300	2000	0308-753578
3	Hannah Zoe	Mianwali	23	76000	6500	900	0333-753957
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- **Not equal to 2000:**



```
68
69 •  SELECT
70      *
71    FROM
72      customer
73    WHERE
74     Payment_Amount != 2000;
75
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
1	Sumaira Safeer	Islamabad	34	4300	1550	450	0333-754468
3	Hannah Zoe	Mianwali	23	76000	6500	900	0333-753957
4	Saqib Hassan	Lahore	65	5600	23500	54500	0344-863579
5	Tayyaba Khan	Multan	87	3400	4500	5400	0300-765367
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Task 6:

Display customer id, customer name, customer city and grade where customer city is New York
and grade is 2.

```
80  SELECT
81    Customer_ID,
82    Customer_Name,
83    Customer_City,
84    Grade
85  FROM
86    customer
87  WHERE
88    Customer_City = 'New York' AND Grade = 2;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conter

Customer_ID	Customer_Name	Customer_City	Grade
2	Aazan Zarsham	New York	2
NULL	NULL	NULL	NULL

Task 7:

Display all records of the customer where grade is not greater than 2.

```
90
91  SELECT
92    *
93  FROM
94    customer
95  WHERE
96    Grade <= 2;
97
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
2	Aazan Zarsham	New York	2	4500	4300	2000	0308-753578
4	Saqib Hassan	New York	1	5600	23500	54500	0344-863579
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Task 8:

Display all from customer where receiving amount is in between the 5000 and 10000.

```
98
99  SELECT
100   *
101 FROM
102   customer
103 WHERE
104   Recieving_Amount BETWEEN 5000 AND 10000;
105
106
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
3	Hannah Zoe	Mianwali	23	76000	6500	900	0333-753957
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Task 9:

Display customer id, customer name, customer city where customer name start with h.

```
107  SELECT
108    Customer_ID,
109    Customer_Name,
110    Customer_City
111  FROM
112    customer
113  WHERE
114    Customer_Name LIKE 'H%';
115
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Customer_ID	Customer_Name	Customer_City
3	Hannah Zoe	Mianwali
NULL	NULL	NULL

Task 10:

Display all records where customer city is new York or London.

SET OPERATORS:

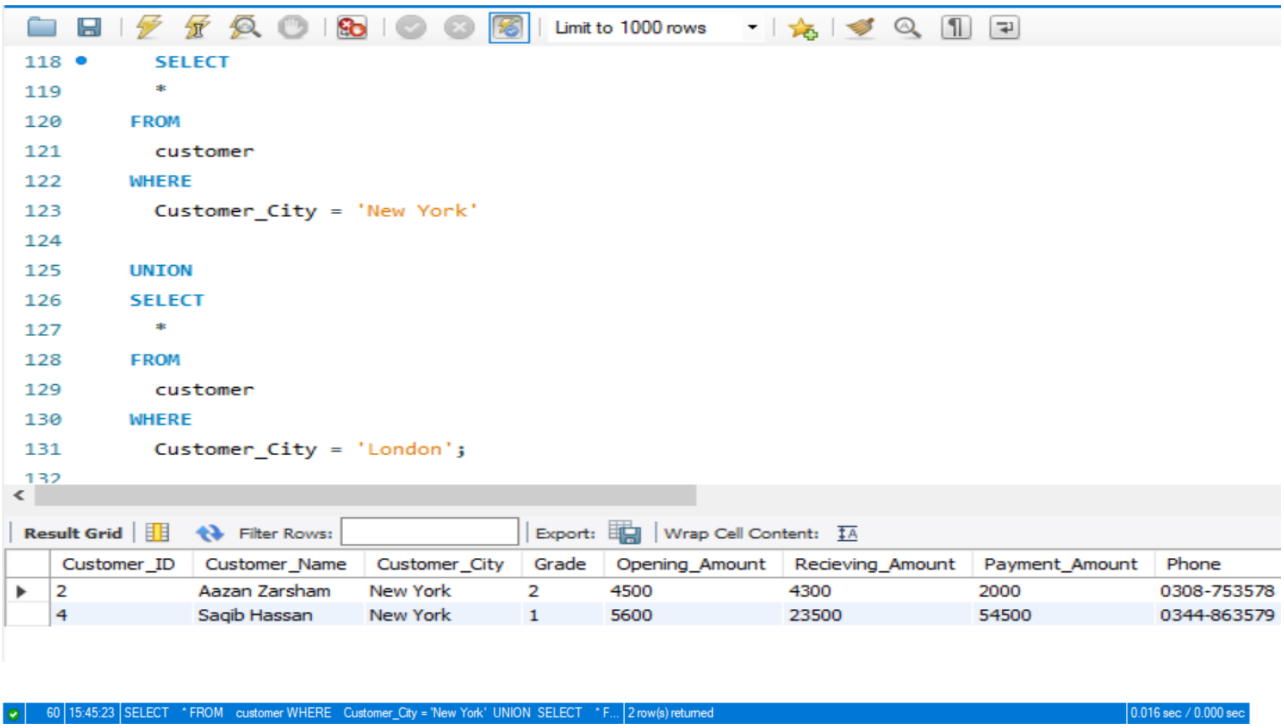
The Set operator combines the result of 2 queries into a single result.

The following are the operators:

Union: Returns all distinct rows selected by both the queries.

Syntax:

SELECT column_name(s) FROM table1 UNION SELECT column_name(s) FROM table2;



```
118    SELECT
119      *
120    FROM
121      customer
122    WHERE
123      Customer_City = 'New York'
124
125    UNION
126    SELECT
127      *
128    FROM
129      customer
130    WHERE
131      Customer_City = 'London';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Customer_City	Grade	Opening_Amount	Recieving_Amount	Payment_Amount	Phone
2	Aazan Zarsham	New York	2	4500	4300	2000	0308-753578
4	Saqib Hassan	New York	1	5600	23500	54500	0344-863579

60 | 15:45:23 | SELECT * FROM customer WHERE Customer_City = 'New York' UNION SELECT * F... | 2 row(s) returned | 0.016 sec / 0.000 sec

Task 11:

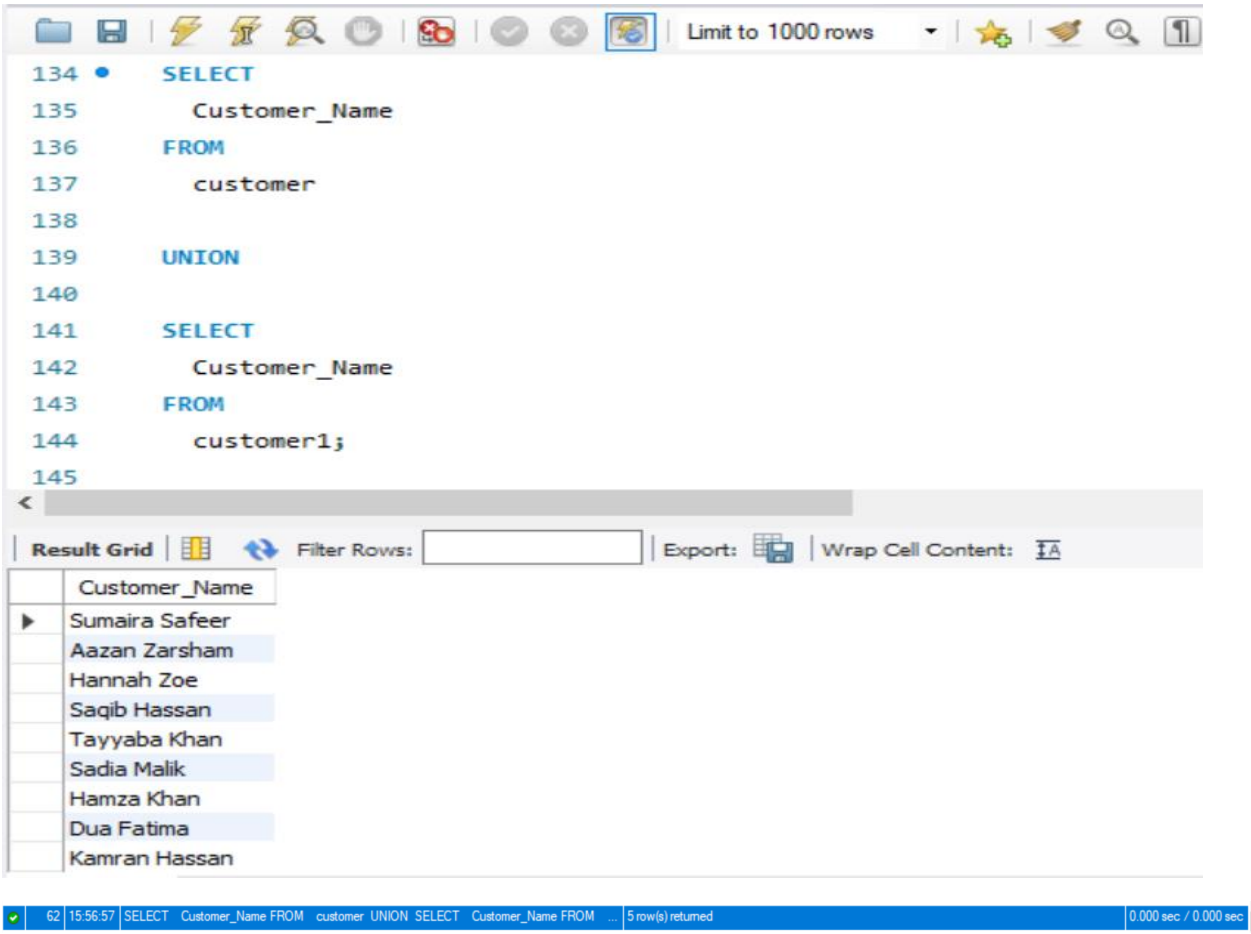
Display customer name from customer and customer1.

Union all:

Returns all rows selected by either query including the duplicates. Display all records from customer and customer1.

Syntax:

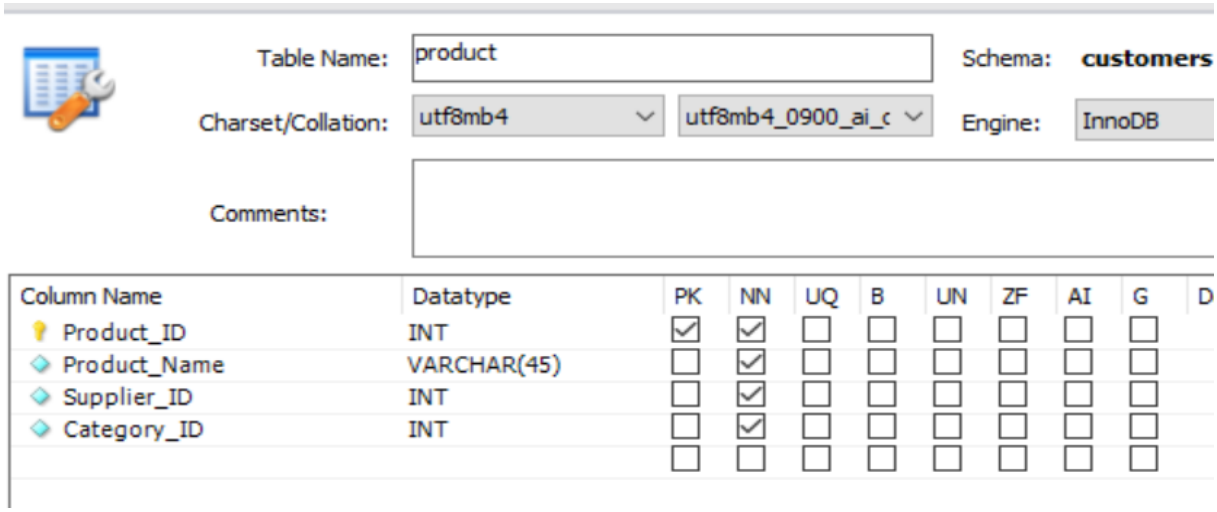
SELECT column_name(s) FROM table1 UNION ALL SELECT column_name(s) FROM table2;


Limit to 1000 rows
134 SELECT
135 Customer_Name
136 FROM
137 customer
138
139 UNION
140
141 SELECT
142 Customer_Name
143 FROM
144 customer1;
145
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Customer_Name
Sumaira Safeer
Aazan Zarsham
Hannah Zoe
Saqib Hassan
Tayyaba Khan
Sadia Malik
Hamza Khan
Dua Fatima
Kamran Hassan
62 15:56:57 SELECT Customer_Name FROM customer UNION SELECT Customer_Name FROM ... 5 row(s) returned
0.000 sec / 0.000 sec

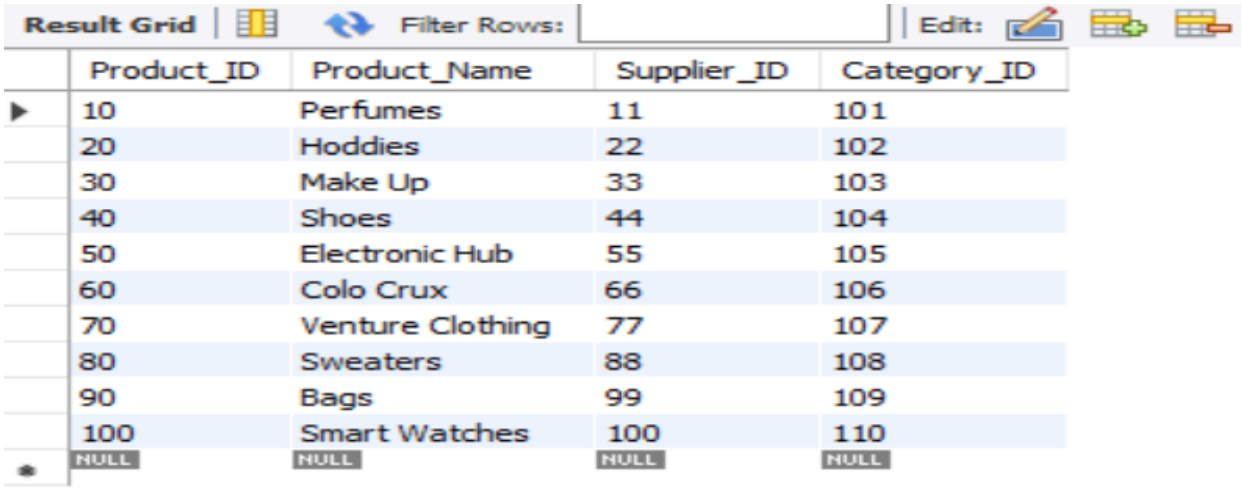
Post lab Tasks:

Consider the following Products Table and OrderDetail.

Products Table:

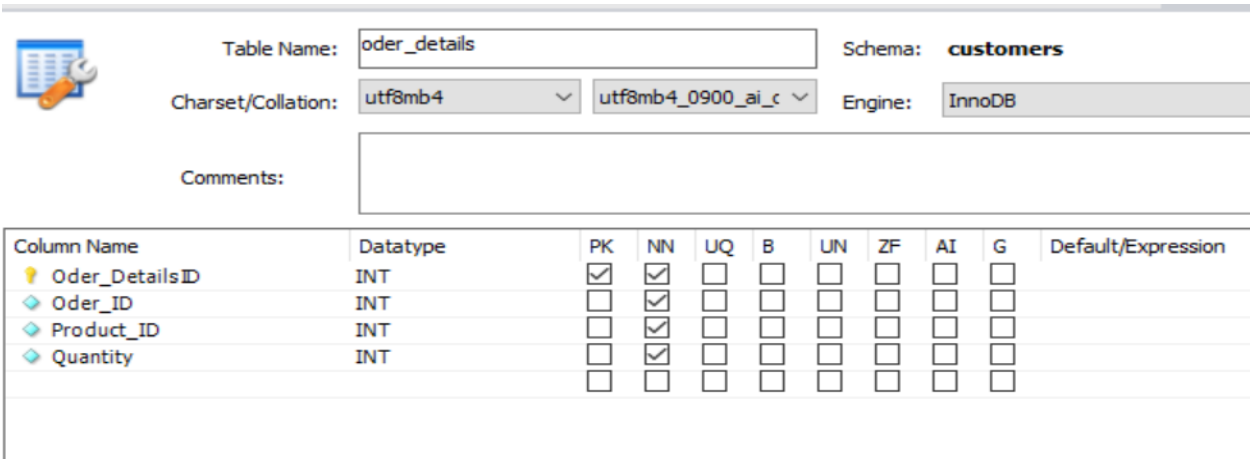


Insertion of data:

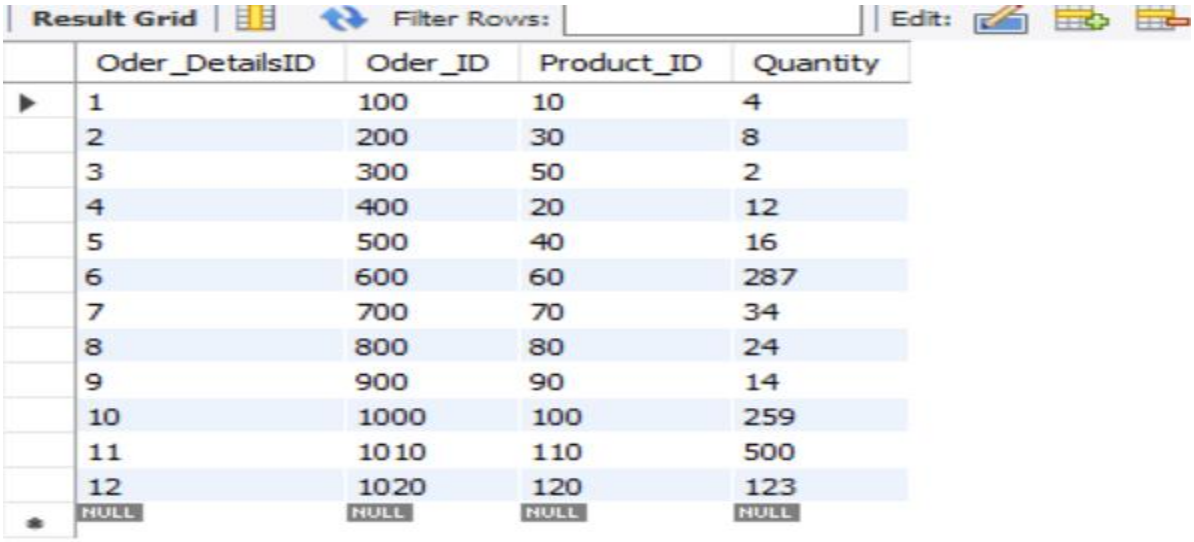


Product_ID	Product_Name	Supplier_ID	Category_ID
10	Perfumes	11	101
20	Hoddies	22	102
30	Make Up	33	103
40	Shoes	44	104
50	Electronic Hub	55	105
60	Colo Crux	66	106
70	Venture Clothing	77	107
80	Sweaters	88	108
90	Bags	99	109
100	Smart Watches	100	110
NULL	NULL	NULL	NULL

OrderDetail Table:

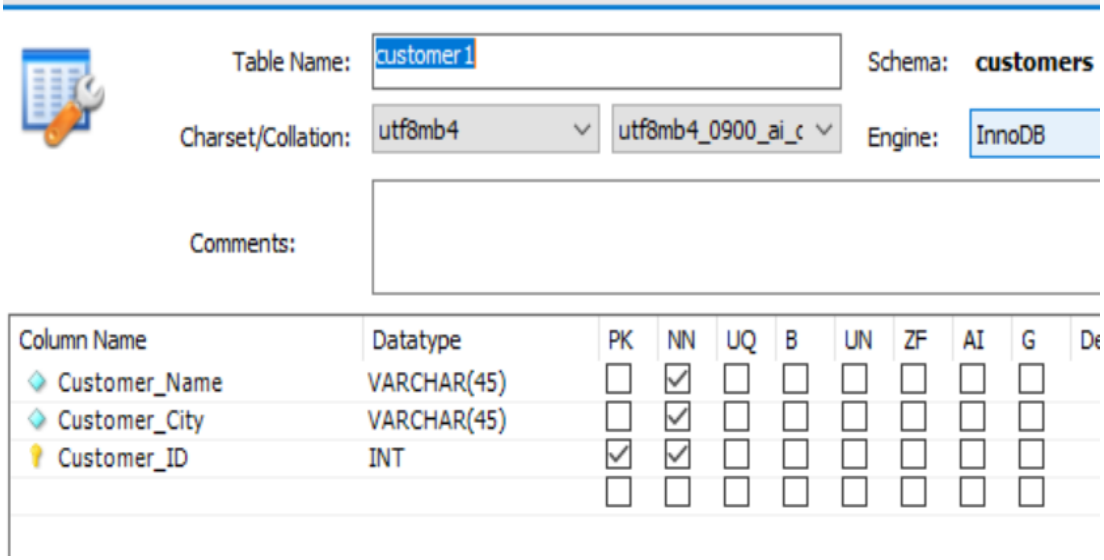


Insertion of data:



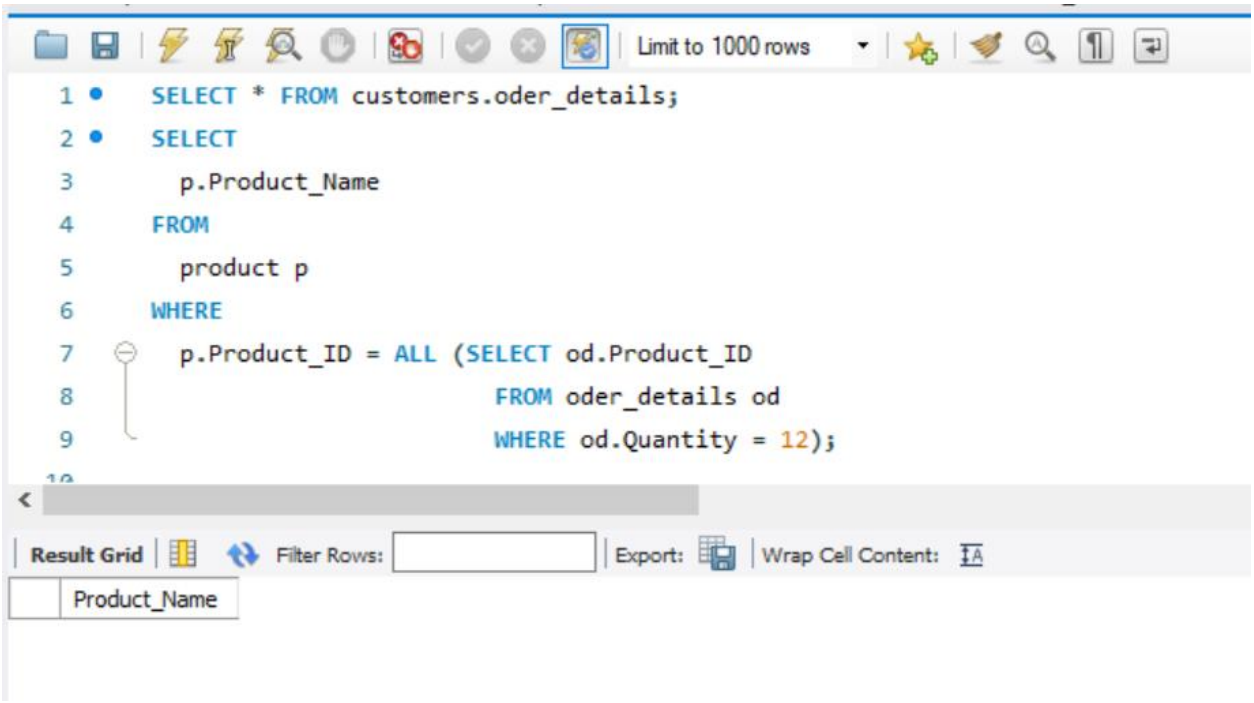
Oder_DetailsID	Oder_ID	Product_ID	Quantity
1	100	10	4
2	200	30	8
3	300	50	2
4	400	20	12
5	500	40	16
6	600	60	287
7	700	70	34
8	800	80	24
9	900	90	14
10	1000	100	259
11	1010	110	500
12	1020	120	123
NULL	NULL	NULL	NULL

Customer Table:



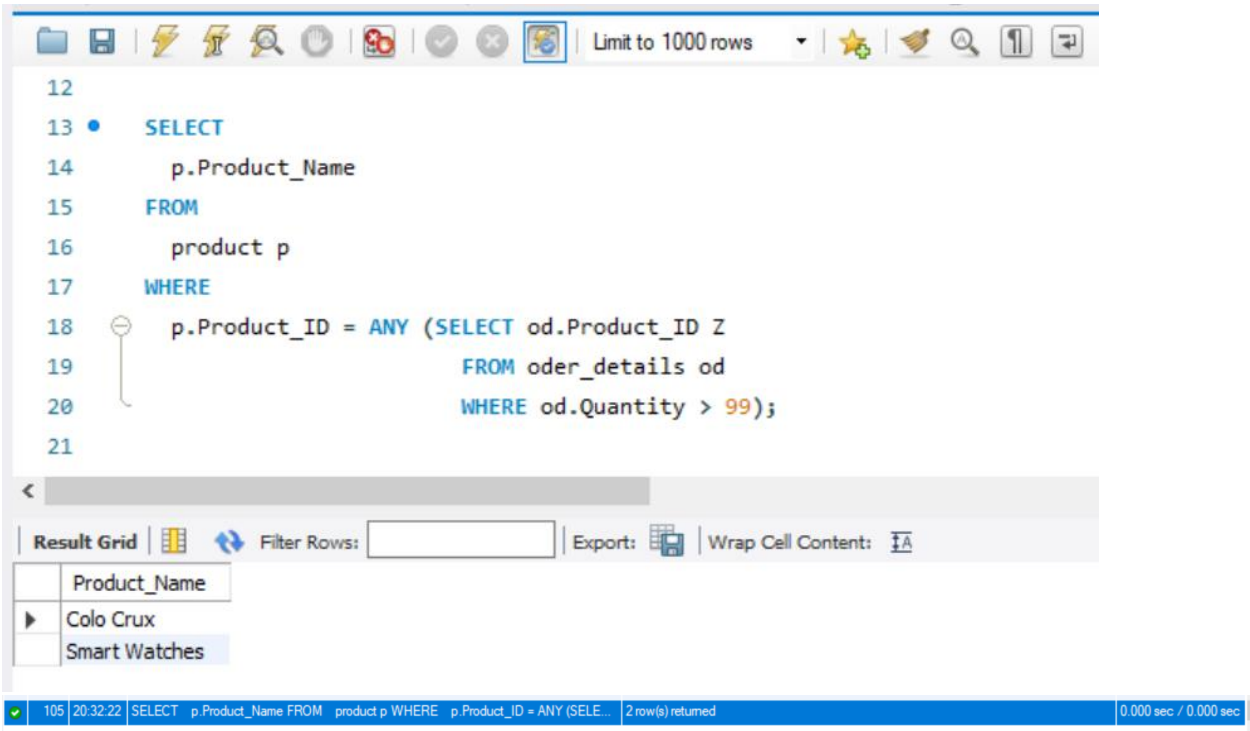
Task 1:

Find the name of the product if all the records in the OrderDetails has Quantity equal to 12 (using All operator).



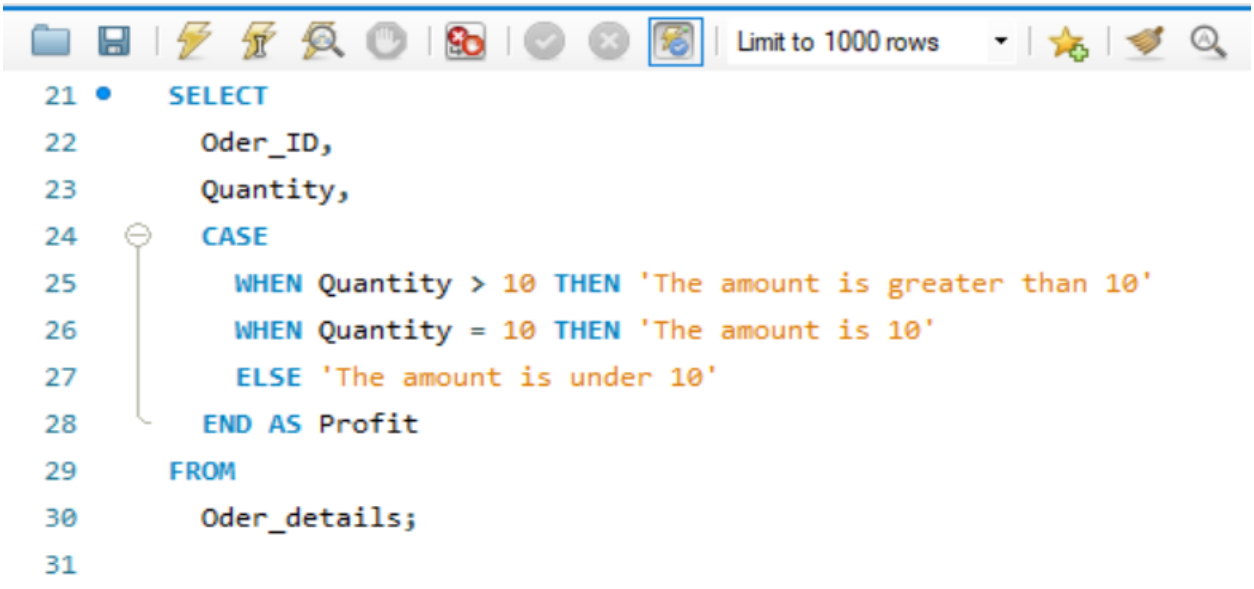
Task 2:

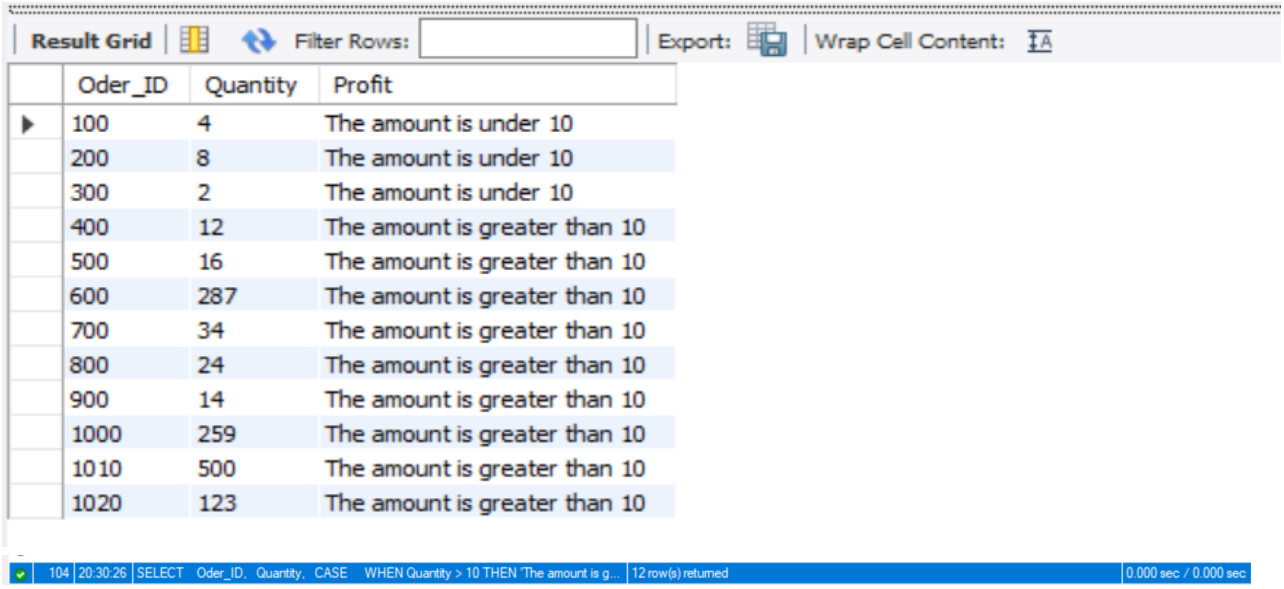
Lists the product names if it finds ANY records in the OrderDetails table that quantity > 99(using ANY operator).



Task 3:

Display OrderID and Quantity, If Quantity is greater than 10 then return “The amount is greater than 10” If Quantity is equal to 10 then return “The amount is10” If Quantity is less than 10 then return “The amount is under 10” The new field name should be ‘Profit’(using case statement).





Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Oder_ID	Quantity	Profit
100	4	The amount is under 10
200	8	The amount is under 10
300	2	The amount is under 10
400	12	The amount is greater than 10
500	16	The amount is greater than 10
600	287	The amount is greater than 10
700	34	The amount is greater than 10
800	24	The amount is greater than 10
900	14	The amount is greater than 10
1000	259	The amount is greater than 10
1010	500	The amount is greater than 10
1020	123	The amount is greater than 10

104 | 20:30:26 | SELECT Oder_ID, Quantity, CASE WHEN Quantity > 10 THEN 'The amount is g... | 12 row(s) returned | 0.000 sec / 0.000 sec